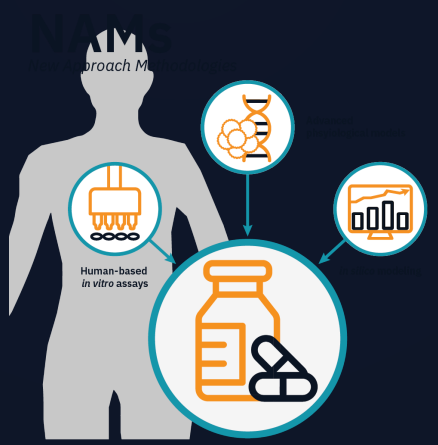
ONDERZOEKSVOORSTEL PITCH · INNOVATIVE TESTING

Multimodal & Multiscale AI in de Toxicologie

1. NEW APPROACH METHODOLOGIES

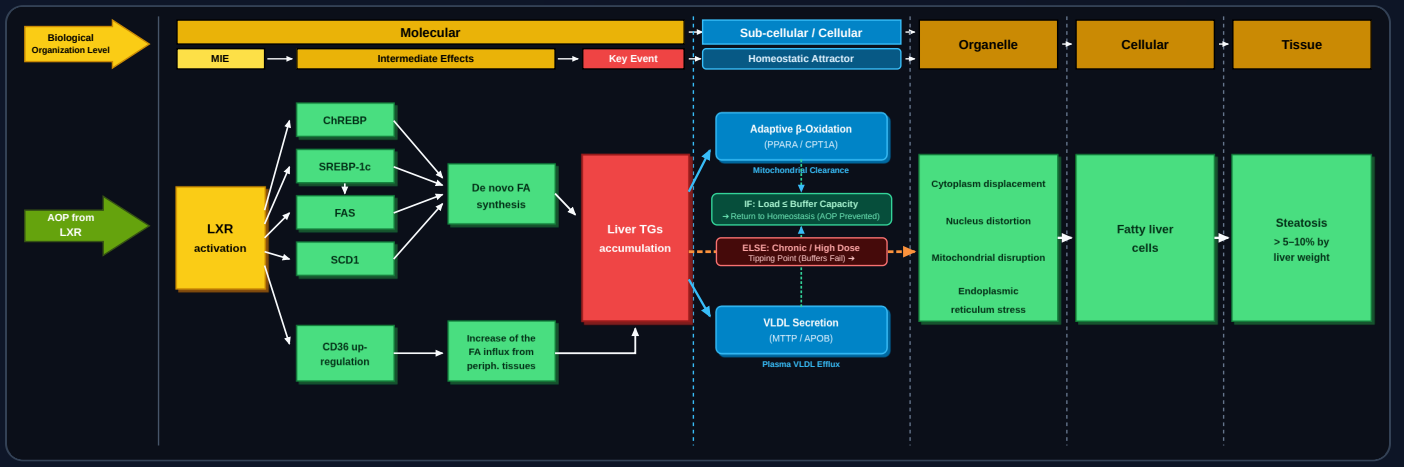


- **Probabilistic Chemical Risk Assessment**
  Accelerating the transition to human-relevant, animal-free biomedical innovation
- **Pillars: (q)AOPs & Disease Maps**
  Mechanistically mapping and explaining *Adverse Outcomes*

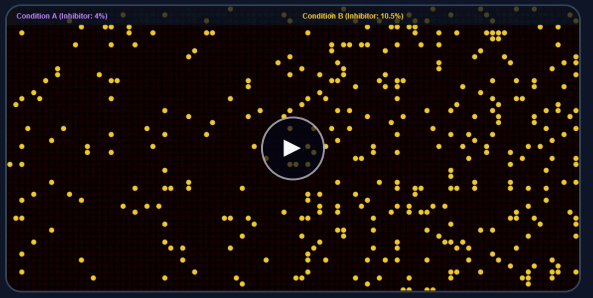
2. EXPANDING ON LINEAR AOPS

From qualitative linear chains to dynamic, quantitative systems predicting $P(z_{homeo}, z_{adversity} \mid \text{context, exposure, dose})$

AOP 34 LXR (MIE 167) → lipogenese → TG-accumulation (KE 291) → steatose



CORTICAL SPREADING DEPRESSION



Breakthrough (left) vs. Propagation Failure (right)
Same MIE: tissue structure determines the fate

SAME MIE, DIFFERENT OUTCOMES

Both tissues initiate with an **identical central potassium perturbation (MIE)**. The outcome only arises through **spatiotemporal emergence**: the local density and buffering capacity of astrocytes (yellow cells) dictate whether the tissue recovers or the wave propagates across the cortex.

scan QR for live simulation

Live Simulator
Scan of klik

Homeostasis or Adversity?

A DEMO CASE a **multimodal & multiscale model** that fuses protein, gene, and chemical data into one representation — and turns it into **probabilities**, not a label.

FOUR DATA SOURCES

Protein–protein interactions
STRING-db v12 · taxon 9606 · PPI
18 genes · mean score 0.81

Gene expression
GTEx v8 · HPA · GEO GSE126848 · HepaRG
18 genes × 3 states

Gene–gene interactions
OECD AOP 34 & 27 + Physiological Maps
20 edges · 14 + / 6 − · [19,19]

Chemical information
SMILES + dose (µM) · Morgan FP 1024 bit
12 compounds · 3 MIE classes

Multi-modal & Multi-scale AI

Graphormer 47.7M + ToxFormer 25.4M

future work

P(MIE | structure)

Chance that a compound induces a specific MIE

MIE 167 · LXRα | MIE 41 · BSEP

leave-one-out
8/12 correct
mean P 0.39
range 0.18–0.51
uncertainty is measurable →

	no MIE	167	41
no MIE	3	0	1
167	0	2	2
41	0	1	3

rows = true · cols = LOOCV

P(z_{homeo} $z_{adversity}$ | context, exposure, dose)

Probability that the hepatocyte tips to $z_{adversity}$ instead of returning to z_{homeo}, given context, exposure and dose

z_{homeo} | $z_{steatosis}$ | $z_{cholestasis}$

Sources: OECD AOP 34 · OECD AOP 27 · STRING-db v12 · GTEx v8 / HPA · GEO GSE126848 · HepaRG (Rodrigues 2018)

6 open sources · 18 genes · 12 compounds · 73.1M parameters